



SEPTEMBER 19, 2025

External Advisory Board Meeting

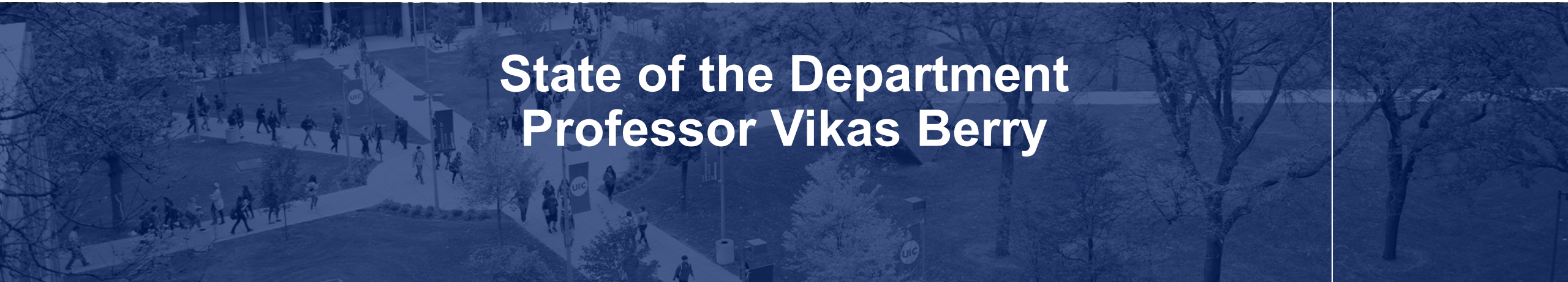


UNIVERSITY OF
ILLINOIS CHICAGO

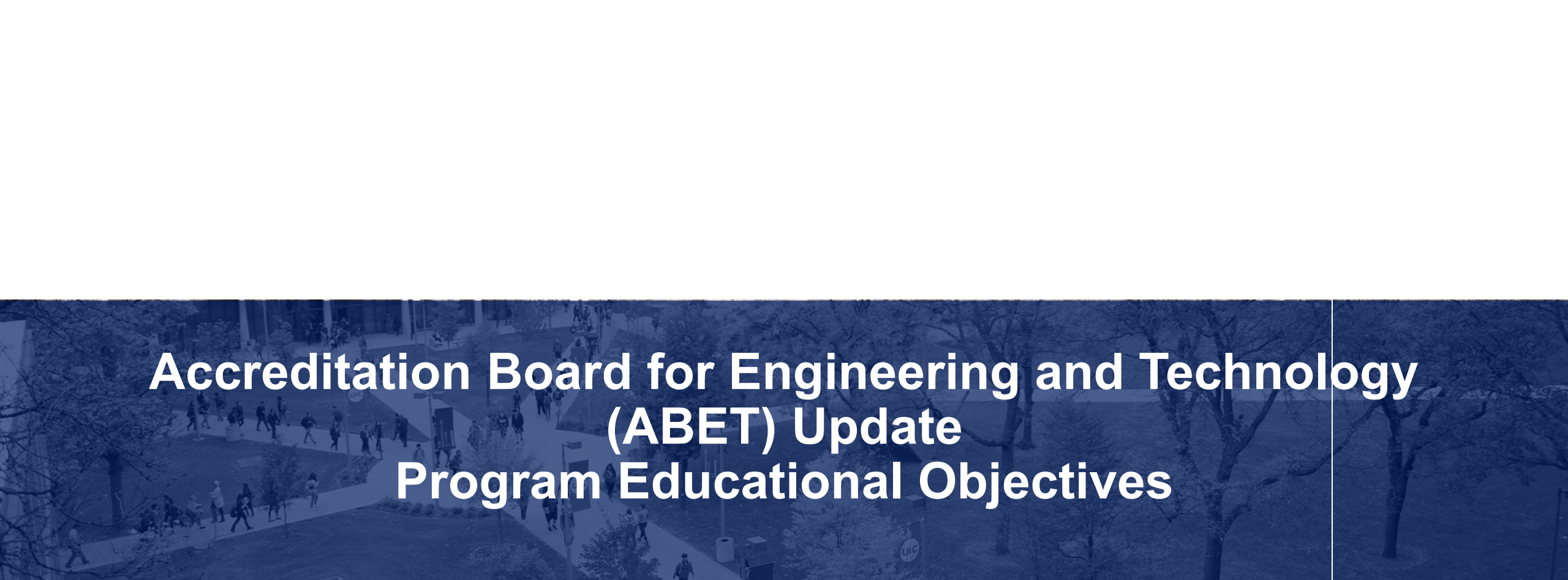
Today's Agenda

Welcome Advisory Board Members

September 19, 2025	EIB Room 224
9:00 a.m. – 9:30 a.m.	Arrival & Check-in with refreshments
9:30 a.m.-9:45 a.m.	Welcome Remarks Alan Zdunek, Clinical Assoc. Professor & Director of Undergraduate Studies
9:45 a.m.-10:15 a.m.	State of the Department Vikas Berry, Professor & Department Head
10:15 a.m.-11:00 a.m.	Undergraduate Curriculum ABET update & Program objectives discussion Alan Zdunek
11:00 a.m.-11:15 a.m.	Break
11:15 a.m.-12:00 p.m.	Graduate Curriculum Brian Chaplin, Professor & Director of Graduate Studies
12:15 p.m.-12:30 p.m.	Department needs and Wrap-up
12:30 p.m. -2:00 p.m.	Lunch & Networking
2:00 p.m.-2:30 p.m.	Building Tour



State of the Department Professor Vikas Berry



Accreditation Board for Engineering and Technology (ABET) Update Program Educational Objectives



UPCOMING ABET REVIEW FALL 2026

LAST REVIEW:
OCTOBER 2000

CHEMICAL
ENGINEERING
PASSED WITH NO
CONCERNS,
WEAKNESSES OR
DEFICIENCIES

Request for Evaluation: December 2026

Current Status:

- (1) Gathering program, college, faculty, facility and course information.
- (2) Performing course assessments, continuous improvement reports
- (3) Self-Study report to be completed by July 1, 2026
- (4) CHE Online Presentation: Summer 2026

More details at next EAB Meeting

file:///E:/Documents/ABET_2020-Complete/ABET%202020%20Presentation%20System/a01b01.htm



ABET Accreditation Review- Fall 2026

8 ABET CRITERIA FORM THE BASIS OF THE ASSESSMENT:

Self-Study Report

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A Review of Current Program Educational Objectives

PEO	Category	Description
1	Depth	In engineering practice, advanced study or original research, UIC Chemical Engineering bachelor's degree graduates will be effective in applying fundamental principles, scientific knowledge, rigorous analysis, creative design, and conceptual innovation.
2	Breadth	Based upon mastery of engineering in a broad, societal context, UIC Chemical Engineering graduates will have successful careers in the public or private sectors, or in the pursuit of graduate education.
3	Professionalism and Service	In the service of industry, government, the engineering profession and society at large, graduates will function effectively in the complex modern work environment with clear communication, responsible teamwork, and high standards of ethics, professionalism, safety and protection of the environment.
4	Life-Long Learning and Leadership	Based upon a rigorous undergraduate program that is innovative, challenging, open and supportive, graduates will enhance their skills and knowledge through life-long learning and demonstrate professional leadership.

Comments?



Undergraduate Curriculum Overview Student Experience Program Improvement



Program Overview

BS Degree: 188 students Fall25
Increase from Fall24 (165)

39% Female

31% Hispanic

20% Asian

9% International

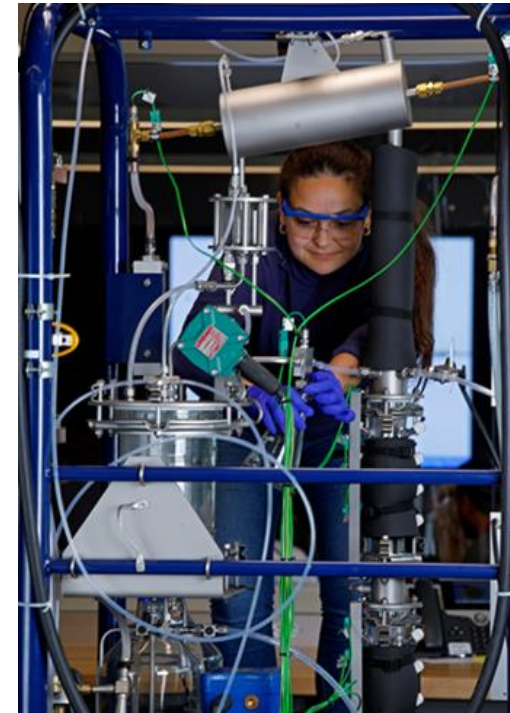
6% Black/African

5% Multi-race

Small class size:

Private-school personal attention
at affordable public-university
tuition rates

We are a Welcoming Community!





ChE Curriculum:

CHE 150 Climate Engineering:
General Education course offered to Non-ChE majors

Five concentrations added Fall 2021

Core Courses

Thermodynamics-Based Courses

ChE 201 – Introduction to thermodynamics
ChE 301 – Chemical Engineering Thermodynamics
ChE 321 – Chemical Reaction Engineering

Process-Oriented Courses

ChE 210 – Material & Energy Balances
ChE 341 – Chemical Process Control
ChE 396 – Senior Design I (Equipment Design)
ChE 397 – Senior Design II (Design teams)

Numerical Analysis

ChE 205- Numerical Methods in ChE

Engineering Education & Development

ENGR100 -Engineering Orientation (Engineering Success Program, Guaranteed Paid Internship Program)
ChE 499- Professional Development Seminar

Transport-Based Courses

ChE 311 – Transport Phenomena I (Fluid Mechanics)
ChE 312 – Transport Phenomena II (Heat & Mass Transfer)
ChE 313 – Transport Phenomena III (Separations)

Unit Operations Laboratory

ChE 381 – Chem Eng Lab I
ChE 382 – Chem Eng Lab II

Molecular Engineering Courses

CHE 230 Molecular Engineering in ChE
CHE 330 Polymer Science

ChE 400-level Elective

CHE 392:
Undergraduate Research

Also Offered in Summer

Six Concentrations (15% of ChE students):

Biochemical Engineering	9 students
Process Automation	6
Energy & Environment	4
Nanotechnology	3
Polymers & Molecular Engineering	3
Entrepreneurship	0



Technical Electives

New permanent
course: CHE 426

New CHE 494
courses

CHE 410 Transport Phenomena
CHE 413 Introduction to Flow in Porous Media

CHE 421 Combustion Engineering

CHE 422 Biochemical Engineering

CHE 423 Catalytic Reaction Engineering

CHE 425 Nanotechnology for Pharmaceutical Applications

★ **CHE 426 Biopharmaceutical Process Development and Manufacturing**

CHE 427 Entrepreneurship in Engineering

CHE 433 Process Simulation w/Aspen+

CHE 438 Computational Molecular Modeling

CHE 440 Non-Newtonian Fluids

CHE 441 Computer Applications in Chemical Engineering

CHE 451 Renewable Energy Technology

CHE 450 Air Pollution

CHE 453 Electrochemistry

CHE 454 Molecular & Macromolecular Engineering

CHE 455 Nanoscale Systems in Chemical Engineering

CHE 456. Fundamentals and Design of Microelectronics Processes

CHE 457 Colloidal and Interfacial Phenomena

CHE 494 Molecular Gastronomy and Engineering

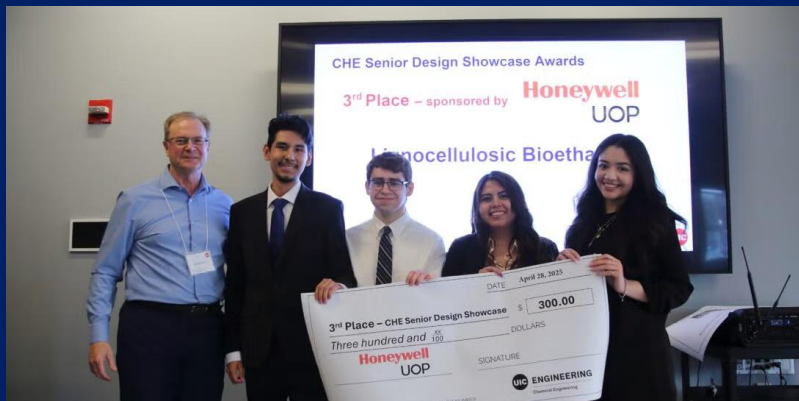
CHE 494 Introduction Chemical Research Methods in Chemical Engineering

CHE 494 Battery and Fuel Cell Development

CHE 494 Clean Energy & the Electric Grid.....?

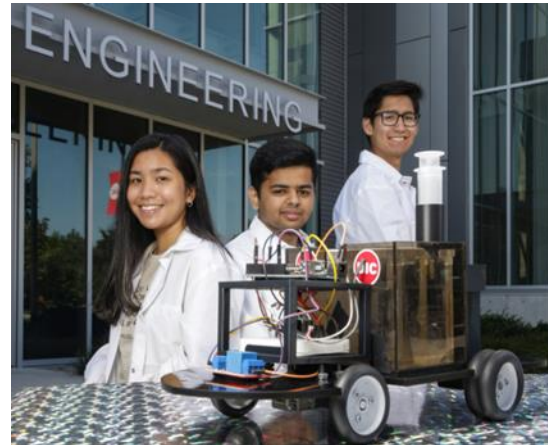
Courses in Red
are Concentration
Electives

Student Experience



STUDENT ACTIVITIES:

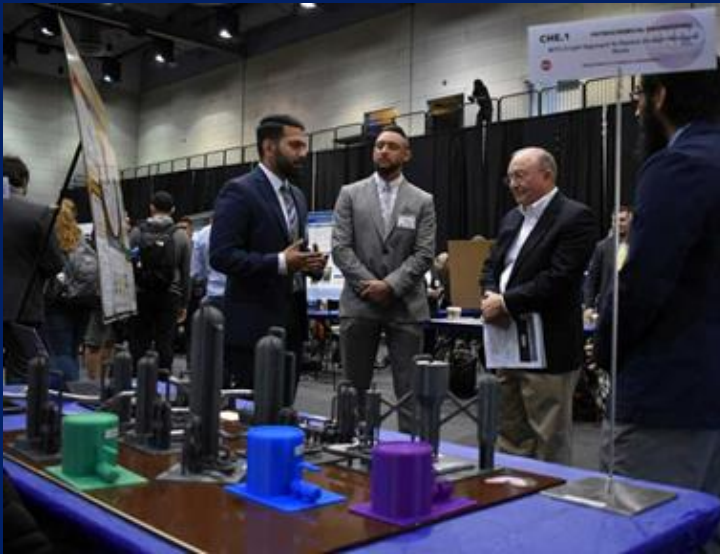
- ACTIVE AICHE STUDENT CHAPTER
- ATTEND NATIONAL AICHE MEETINGS
- HOSTED NORTH-CENTRAL REGIONAL CONFERENCE 2022
- CHEME CAR, CHEME CUBE COMPETITIONS



- CHE DEPARTMENT ACTION TEAM: STUDENTS/FACULTY/STAFF
- GPIP, INTERNSHIPS, CO-OPS
- CO-AUTHORS ON RESEARCH PUBLICATIONS
- PARTICIPATION IN OTHER UIC STUDENT ORGANIZATIONS
- SWE AMBASSADORS
- HONORS COLLEGE
- UNDERGRADUATE MENTORS TO ENGR100



Student Experience



STUDENT/INDUSTRY/ALUMNI INTERACTION

CHE ALUMNI AND INDUSTRIAL PARTNERS PROVIDE STRONG SUPPORT TO OUR STUDENTS AND DEPARTMENT ACTIVITIES:

- 499 SEMINAR SPEAKERS
- 396/397 SENIOR DESIGN MENTORS, EXPO JUDGES
- AICHE STUDENT CHAPTER ACTIVITIES/TOURS
- AICHE CHICAGO SECTION MCCORMACK AWARD
- AICHE CHICAGO SECTION STUDENT POSTER SESSION
- FEEDBACK ON CURRICULUM IMPROVEMENTS

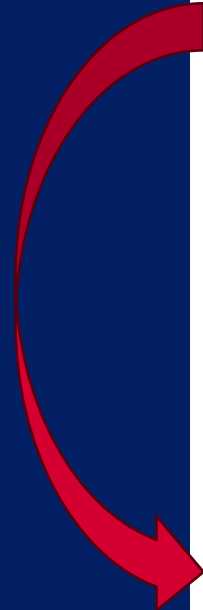
~750 ALUMNI IN CONTACT LIST: 2012 TO 2024 UPDATED EVERY SPRING:

UIC CHEMICAL ENGINEERING LINKEDIN GROUP



Curriculum Improvement

Recent curriculum changes



PROGRAM REVISIONS:

(1) NEW COURSES

CHE 230: MOLECULAR SYSTEMS IN CHE
REQUIRED IN BS DEGREE

CHE 330: POLYMER SCIENCE

CHE 150: CLIMATE ENGINEERING FOR
GLOBAL WARMING

GENERAL EDUCATION COURSE OFFERED TO
NON-CHE MAJORS

(2) ADDED SELECTIVE OPTION TO DEGREE

CHE 330 POLYMER SCIENCE **OR**

CHEM 346 PHYSICAL CHEMISTRY II



Curriculum Improvement: Add CHE 101 to curriculum

Approved by the COE
Education Policy Committee
(EPC), April 2025

CHE 101: Introduction to Chemical Engineering and Sustainability

Overview of the principles and unit operations of chemical engineering as practiced in traditional industries and in sustainable technologies.

Chemical engineering/unit operations, ChE traditional processes, case studies

Five sustainable themes (modules):
Energy, Health, Food, Air, Water

Students will explore the challenges in these areas and the role of chemical engineering in developing sustainable products and processes.

Modules will include a guest speaker (faculty or external)



Possible program changes to add CHE 101 to curriculum

(1) ADD CHE 101: INTRODUCTION TO SUSTAINABLE CHEMICAL ENGINEERING TO CORE CURRICULUM (+3 HOURS)

THIS WILL REQUIRE REMOVING A COURSE FROM OUR CURRICULUM:

SOME IDEAS:

CREATE A NEW SELECTIVE OPTION:

(1) CHE 230 MOLECULAR SYSTEMS IN CHEMICAL ENGINEERING **OR** CME 260 PROPERTIES OF MATERIALS (-3 HOURS)

(2) PHYSICAL CHEMISTRY I OR ANALYTICAL CHEMISTRY (- 3 HOURS)



Future Curriculum Improvement: Chemical Engineering & Data Science

CHEMICAL ENGINEERING AND DATA SCIENCE

HOW CAN WE ADD DATA SCIENCE TO OUR
CURRICULUM?

(1)

BS IN CHE WITH DATA SCIENCE
CONCENTRATION

(UIUC IS IMPLEMENTING A SIMILAR
SPECIALIZATION)

(2)

BS IN DATA SCIENCE WITH CHE
CONCENTRATION

(UIC COMPUTER SCIENCE DEPARTMENT HAS
SEVERAL PROGRAMS WITH OTHER MAJORS)





Program Improvement

Future changes

**(1) ADD DATA SCIENCE COURSES TO
CURRICULUM**

**(2) REQUIRE TWO, 400-LEVEL TECHNICAL
ELECTIVES IN CORE BS DEGREE**

**THIS WILL REQUIRE REMOVING COURSES
FROM OUR CURRICULUM:**

Program Branding for Recruitment Efforts



EMPHASIZE CHEMICAL ENGINEERS' ROLE IN SUSTAINABILITY:

UNLOCK THE FUTURE WITH CHEMICAL ENGINEERING!

Imagine being at the forefront of groundbreaking innovations. Chemical engineers are transforming the world by creating renewable energy, providing clean water, developing life-saving drugs, designing eco-friendly products, and using AI and nanotechnology to revolutionize industries.

Ready to make a real difference and change the world?
Join us and become part of this amazing journey!

**UIC SPARKFEST (OPEN HOUSE), IGNITE EVENT
HIGH SCHOOL VISITS**

COMMUNITY COLLEGE VISITS

WOMEN IN ENGINEERING SCHOLARSHIP BRUNCH

PERSONALIZED COMMUNICATION TO APPLICANTS

An aerial photograph of a university campus, likely the University of Illinois at Chicago (UIC), is shown with a semi-transparent blue overlay. The image captures a wide pedestrian walkway lined with trees, with many people walking. Several UIC logos are visible on signs and banners along the path. The word "BREAK" is centered in the image in a large, white, sans-serif font.

BREAK



Graduate Program Update to External Advisory Board

Sept 19, 2025

Brian P. Chaplin

Professor

Director of Graduate Studies



UNIVERSITY OF
ILLINOIS CHICAGO



GRADUATE PROGRAM OVERVIEW

Demographics and Recruiting:

- 53 Graduate Students: 42 PhD/11 MS
- 60% Male and 40% Female
- ~19% Domestic students
- International students primarily from India, Iran, and China (decreasing).
- Increasing numbers from Nigeria.
- Recruiting concerns:
 - How do we recruit effectively from other countries?
 - Many international students cannot afford graduate school in US (MS degree).
 - How do we increase domestic student enrollment?



MS CHEMICAL ENGINEERING

Value of CHE MS Degree:

- Historically, most industry jobs in chemical engineering required only a BS.
- MS degrees were not standard industry practice.
- Industrial demand for advanced expertise in biotech, pharma, energy, and sustainability has increased the value of MS degrees since the 2010s.
- Modern MS programs in chemical engineering emphasize specialized skills and prepare graduates for advanced roles, leadership, and higher salaries.
- MS increasingly boosts career mobility and access to interdisciplinary and innovative sectors.



MS CHEMICAL ENGINEERING

MS Degree Options:

- Three formats: two-year, one-year fast track, and 4+1 BS/MS
- All students gain advanced chemical engineering knowledge with customizable coursework and options for thesis, project, or coursework-only paths.
- Core curriculum covers transport phenomena, thermodynamics, separation processes, reaction engineering, and mathematical methods.
- 36 credit hours for all paths, including core courses, electives, and research/project (if applicable).
- We are proposing **MS in Pharmaceutical Engineering**



MS PHARMACEUTICAL ENGINEERING

Program Overview:

- Proposing a new specialized graduate program in biopharmaceutical engineering at UIC
- Builds on existing Chemical Engineering foundations
- Designed for industry demand in Chicago and nationwide
 - Chicago Big Pharma: AbbVie, Abbot Laboratories, Baxter International, Takeda, etc.
 - Local Biotech: Tempus, Emalex Biosciences, CancerIQ



MS PHARMACEUTICAL ENGINEERING

Curriculum Highlights:

- Thesis, Project, or Coursework-Only Options
- 36 semester hours required
- **Core Courses:**
 - Biopharmaceutical Process Development and Manufacturing (4 CH)
 - Particle Technology (4 CH)
 - Advanced Reaction Engineering (4 CH)
 - Pharmaceutical Reaction Engineering (2 CH)
 - ChemE Core (Thermo, Transport, or Math) (4 CH)
- Seminars and electives included for thesis and professional development



MS PHARMACEUTICAL ENGINEERING

Elective Options Examples:

- Nanotechnology in Pharma Applications
- Pharmacokinetics (PHAR 435)
- Biochemical Process
- Pharmaceutical Characterization
- Regulatory Framework in Pharma Industry
- Colloidal and Interfacial Phenomena
- Advanced Drug Delivery Systems (PSCI 518)
- Drug Discovery, Design, and Development (PSCI 501)



MS PHARMACEUTICAL ENGINEERING

Program Rationale:

- Builds on strong UIC Chemical Engineering foundations with added biomanufacturing and biotechnology coursework and research opportunities
- Students gain rigorous training in chemical engineering plus specialized skills in areas like fermentation, downstream processing, drug formulation, and quality control
- Utilizes shared resources, faculty, and courses for cost-effective, efficient program implementation
- Addresses high demand in Chicago's pharmaceutical and biotech industries for specialized engineering talent



MS PHARMACEUTICAL ENGINEERING

Program Rationale:

- Aligns UIC with peer institutions offering similar programs, keeping UIC competitive and attractive to top students
- Expands UIC's academic portfolio, enhances research synergies, and opens interdisciplinary collaboration opportunities
- Prepares graduates for impactful careers in biopharmaceutical engineering within a rapidly growing market
- Can be supported by Industrial Advisory Board input, which guides curriculum, internships, and industry partnerships



MS PHARMACEUTICAL ENGINEERING

Schools with Similar Programs:

Illinois Institute of Technology:

<https://catalog.iit.edu/graduate/colleges/engineering/chbe/master-pharmaceutical-engineering/>

Carnegie Mellon:

<https://www.cmu.edu/ms-biotech-pharma/curriculum/index.html>

Northeastern:

<https://catalog.northeastern.edu/graduate/engineering/chemical/pharmaceutical-engineering-ms/#programrequirementscontainer>

Rutgers University:

<https://pharmeng.rutgers.edu/courses/core-courses/>

New Jersey Institute of Technology (NJIT):

<https://catalog.njit.edu/graduate/newark-college-engineering/chemical-materials-engineering/pharmaceutical-ms/>



DISCUSSION QUESTIONS

MS Program Value and Industry Needs

- What specialized skills or coursework do you view as most critical for MS graduates entering the biotech, pharma, energy, or sustainability sectors?
- Are there industry trends or emerging areas that should be reflected in our MS and proposed Pharmaceutical Engineering curricula?
- How can our curriculum development stay ahead of industry changes to maintain the employability of our graduates?

New MS in Pharmaceutical Engineering

- How can we best leverage existing local and national partnerships to enhance experiential learning, research, or internships for this new program?
- What opportunities do you see for interdisciplinary collaboration, either within UIC or with external partners, to make this program more impactful?
- What support or input can the advisory board provide to ensure the successful launch and sustained relevance of this new MS specialization?

Broader Curriculum and Program Feedback

- How can we further involve industry and alumni in providing feedback on curriculum content, delivery, and outcomes assessment?
- What key performance indicators should the board or department track to measure the success of the MS and Pharmaceutical Engineering programs?



Department Needs/Wrap-Up

Next Meeting: Summer 2026

Extra Slides



Concentration requirements

Energy & Environment

CHE 330 Polymer Science

CHE 451 Renewable Energy Technology

CME 322 Environmental Engineering

Choose One Course:

CHE 450 Air Pollution or

CHE 453 Electrochemistry or

400-level course approved by DUGS or

CHE 392 Undergraduate Research

Biochemical Engineering

CHE 422 Biochemical Engineering

CHE 494 Biopharmaceutical Process Engineering

Choose two courses:

BIOS 350 General Microbiology

BIOS 351 Microbiology Laboratory

CHEM 352 Introductory Biochemistry

CHEM 452 Biochemistry I



Concentration requirements

Process Automation

CHE 433 Process Simulation w/Aspen+

Choose Two Courses:

ECON 120 Principles of Microeconomics

IE 201 Financial Engineering

MATH 310 Applied Linear Algebra

MATH 419 Models in Applied Mathematics

Entrepreneurship

CHE 427 Entrepreneurship in Engineering

Choose one course:

ENTR 310 Introduction to Entrepreneurship

ENTR 200 Survey of Entrepreneurship

400-level course approved by the chemical engineering department's director of undergraduate studies

ENTR 445 New Venture Planning



Concentrations Requirements

Polymers & Molecular Engineering

CHE 330 Polymer Science

CHE 454 Molecular & Macromolecular Engineering

Choose One Course:

CHE 425 Nanotechnology for Pharmaceutical Applications

CHE 438 Computational Molecular Modeling

CHE 440 Non-Newtonian Fluids

CHE 453 Electrochemistry

CHE 455 Nanoscale Systems in Chemical Engineering

CHE 457 Colloidal and Interfacial Phenomena

400-level course approved by the Director of Undergraduate Studies

CHE 392 Undergraduate research

Choose One Course:

PHAR 422 Fundamentals of Drug Action

PHAR 423 Biomedical Chemistry

PHYS 450/BIOE 450 Molecular Biophysics of the Cell

CHEM 452 Biochemistry I

CHEM 458/BIOS 458 Biotechnology and Drug Discovery

PHYS 461 Thermal and Statistical Physics

BIOE 485 Biosensors



Concentration requirements

Nanotechnology

CHE 330 Polymer Science

CHE 455 Nanoscale Systems in Chemical Engineering

Choose one course:

CHE 425 Nanotechnology for Pharmaceutical Applications

CHE 457 Colloidal and Interfacial Phenomena

400-level course approved by the Director of Undergraduate Studies

CHE 392 Undergraduate research on a topic thematically linked with polymer science or molecular engineering

Choose one course:

ECE 346 Solid State Device Theory

ME 418 Transport Phenomena in Nanotechnology

PHYS 431 Modern Physics: Condensed Matter

ECE 440 Nanoelectronics

ECE 449/ME 449 Microdevices and Micromachining Technology

CHEM 458/BIOS 458. Biotechnology and Drug Discovery

BIOE 485. Nanobiosensors